

Real time Bridge between Audio Encoder and LLM

Problem & Solution

Problem: Audio encoders produce dense frame sequences (tens of frames/sec). Feeding frames directly to small LLMs explodes context length and latency.

Solution: A **streaming adapter network** that compresses continuous audio into **low-bitrate, append-only audio tokens** (1-3 tokens/sec) that a frozen small LLM can consume in real-time.

Core Mechanism

Chunked Streaming Pipeline:

1. Audio split into overlapping windows (0.8s window / 0.4s stride)
2. Frozen audio encoder (Whisper) $\rightarrow$ frame features $F \in \mathbb{R}^{(T \times D_{\text{enc}})}$
3. **Trainable adapter** compresses each window $\rightarrow m$ tokens (1-4 tokens/window)
4. Tokens appended to LLM context (KV-cache friendly)
5. LLM generates early and continues as more tokens arrive

Target: ~1-3 tokens/sec (2 minutes $\approx$ 240 tokens vs. 3000+ frames)

Architecture (4 Components)

1. **Frozen Audio Encoder** (Whisper): Raw audio $\rightarrow$ frame features
2. **Streaming Adapter Network** (trainable):
 - Q-Former style cross-attention resampler
 - Query vectors $Q \in \mathbb{R}^{(m \times D_q)}$ (learnable, $m=1-4$)
 - Cross-attention: $\text{Attn}(Q, F)$ compresses frames $\rightarrow$ tokens
 - Stability buffer for temporal consistency
 - Optional adaptive token rate controller

3. **Early-Commit Gate** (optional, trainable): Decides when to start generating vs. keep listening
4. **Frozen LLM** (small LLaMA/Phi/Mistral): Consumes accumulated tokens

Research Novelties

A) Token Efficiency: Adaptive token rate—fewer tokens for clear speech, more for noisy/ambiguous audio

B) Stability: Stability objective prevents token "thrashing" as new windows arrive (reduces "oops, I meant" behavior)

- Temporal consistency loss: $L_{\text{stability}} = \sum_t \text{MSE}(Z_t, Z_{t-1})$
- Prefix consistency: Penalizes changes in LLM predictions when new tokens arrive

C) Early-Commit Policy: Trainable gate learns latency-accuracy tradeoff (when to start responding)

Training Path (3-Stage)

Stage 1: Audio-Text Alignment

- Contrastive loss: Align adapter tokens with text embeddings
- Dataset: Large-scale audio-transcript pairs (LibriSpeech, Common Voice)
- Loss: $L_{\text{align}} + \lambda_{\text{stability}} \cdot L_{\text{stability}}$

Stage 2: Content Preservation

- ASR distillation: Ensure tokens preserve lexical meaning
- Loss: $L_{\text{asr}} + \lambda_{\text{align}} \cdot L_{\text{align}} + \lambda_{\text{stability}} \cdot L_{\text{stability}} + \lambda_{\text{sparse}} \cdot L_{\text{sparse}}$

Stage 3: Task Distillation

- Knowledge distillation from teacher text LLM
- Train system to answer directly from audio tokens
- Loss:
 $L_{\text{task}} + \lambda_{\text{asr}} \cdot L_{\text{asr}} + \lambda_{\text{stability}} \cdot L_{\text{stability}} + \lambda_{\text{rate}} \cdot L_{\text{rate}} + \lambda_{\text{gate}} \cdot L_{\text{gate}}$

Key: Only adapter (and optional LoRA) trains; encoder and LLM remain frozen.

Loss Functions

- **Alignment:** Contrastive audio-text embedding similarity
- **Stability:** Temporal consistency + prefix consistency + revision penalty
- **Token Efficiency:** Sparsity regularization + rate control loss
- **Task Performance:** KL divergence from teacher LLM distributions

Optimization Techniques

- **Optimizer:** AdamW (LR: $1e-4$ to $5e-4$, stage-dependent)
- **Memory:** Mixed precision (BF16/FP16), gradient checkpointing for adapter only
- **Stability:** EMA of adapter weights, gradient clipping ($\text{max_norm}=1.0$)
- **Batch Construction:** Chunked batching with curriculum learning (start short, increase length)

Evaluation Metrics (3 Axes)

1. **Task Quality:** QA/intent/summarization accuracy
2. **Latency:** First-token latency, time-to-correct-answer
3. **Stability:** Revision rate, prefix-consistency (how much early output changes)

Theoretical Foundation

- **Information Bottleneck:** Minimize $I(\text{Audio}; \text{Tokens})$ while maximizing $I(\text{Tokens}; \text{Task})$
- **Compression:** 25:1 reduction (T frames $\rightarrow m$ tokens, e.g., $50 \rightarrow 2$)
- **Prefix Consistency:** $P(y \mid Z_{\{1:t\}}) \approx P(y \mid Z_{\{1:t+k\}})$ for stability
- **Rate-Distortion:** Target $R = m/t$ while maintaining $I(\text{Audio}; \text{Tokens}) \approx I(\text{Audio}; \text{Text})$

Related Work

- **SALMONN**: Window-level Q-Former (we extend to streaming stability)
- **BLIP-2/Flamingo**: Q-Former/Perceiver resampler concept (we adapt to streaming audio)
- **MoChA**: Monotonic attention for online processing
- **Moshi/Mini-Omni/LLaMA-Omni**: Low-latency speech interaction benchmarks